Week 3 Report: Supervised Models (KNN, Decision Tree, Logistic Regression)

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GitHub repository: https://github.com/MHassanEjaz/rag-project

1. Dataset

arXiv paper metadata (Hugging Face mirror of the Kaggle arXiv dataset), the same data as Week 2. 19,979 rows, 11 input columns (7 numeric, 4 categorical; 78 features after encoding and scaling). Target: field (the main subject area of a paper), 10 classes, so this is multi-class classification. Train 15,983 rows, test 3,996 rows (80/20, stratified, random_state=42).

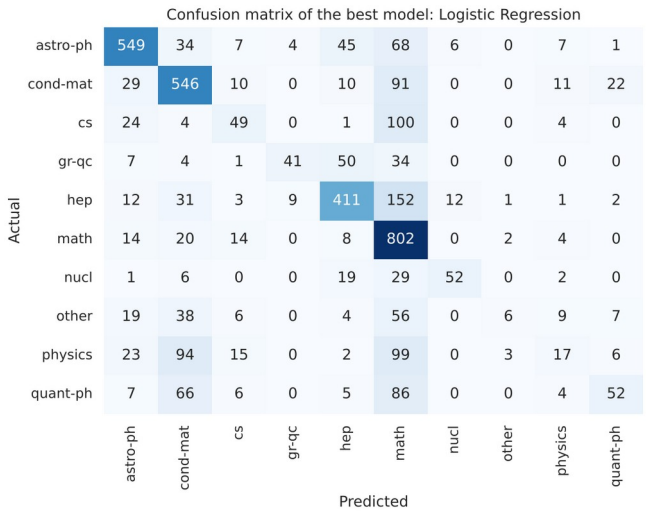
Class balance: the data is not balanced. The biggest class ('math', 4,321 rows) is 7.9 times bigger than the smallest ('nucl', 544 rows). All counts: math 4,321, astro-ph 3,604, cond-mat 3,595, hep 3,169, physics 1,297, quant-ph 1,131, cs 908, other 727, gr-qc 683, nucl 544.

2. Results of the three models (test set)

Model	Accuracy	Precision	Recall	F1	Macro F1
KNN	0.621	0.600	0.621	0.586	0.453
Decision Tree	0.547	0.562	0.547	0.520	0.386
Logistic Regression	**0.632**	**0.629**	**0.632**	**0.596**	**0.470**

Precision, Recall and F1 are weighted averages over the classes; Macro F1 counts every class equally. Best model: Logistic Regression (highest Macro F1, accuracy 63.2%). A model that always predicts the biggest class would get 21.6% accuracy and 0.036 Macro F1. For Logistic Regression accuracy is 0.162 higher than Macro F1, so accuracy hides weak small classes and is misleading on its own.

3. Confusion matrix of the best model (Logistic Regression)



Each row is the true field and each column is what the model predicted. The diagonal shows correct answers: 2,525 of 3,996 test papers (63.2%). Everything off the diagonal is a mistake.

- The most common mistake: real 'hep' papers predicted as 'math' (152 times).
- Weakest classes: 'other' (only 4.1% of its papers found) and 'physics' (6.6%).
- 'math' absorbs the most wrong predictions (715 papers from other fields were labelled 'math'); it is the biggest class, so the model leans towards it when unsure.

4. Best K and best max_depth

- Best K = 15. I tried every odd K from 1 to 15 and chose K by 5-fold cross-validation on the training data (cross-validation accuracy 0.595, test accuracy 0.621), so the test set was not used to make the choice. For K=5 the test accuracy is 59.0%.
- The best K is the largest value I tried, and accuracy was still rising slowly, so a bigger K (above 15) might do a little better.
- Best max_depth = 10, again by cross-validation (cross-validation accuracy 0.529, test accuracy 0.547).
- A very deep tree overfits: with no depth limit the tree grew to depth 52 with 6,251 leaves and reached 99.8% accuracy on the training data but only 52.2% on the test data. It memorises the training papers instead of learning general rules.

5. Which metric matters most

With 10 classes a false positive or false negative is counted per class. In my project the predicted field will be used to filter or route questions to the right papers. A false negative (a paper of the right field is missed) means the answer may lack its real sources, while a false positive (one extra paper let in) can be removed later by the reranker. So recall matters more than precision. Because the classes are unbalanced I follow Macro F1 together with recall, not accuracy alone (63.2% accuracy vs 0.470 Macro F1 for Logistic Regression).

6. What I learned and problems I faced

- Accuracy alone is not enough: the always-biggest-class baseline already gets 21.6%. Macro F1, per-class recall and the confusion matrix show how the small classes are really doing.
- Choosing K and max_depth with cross-validation on the training data keeps the test set clean; using the test set for tuning would make the final score too optimistic.
- The task is multi-class, so precision, recall and F1 need averaging (weighted and macro) and false positives and false negatives must be read per class.
- Linear Regression cannot predict a class, so I used it for a separate task, predicting abstract length in words: R2 = 0.33 and MAE = 41 words, compared with 50 words for always guessing the average.